Ordinal relations: adjacent pair

Inputs (one-hot encoded items, latent rank $A < B < C < D < E$):

$$X = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} \leftarrow \text{token 0: item } B \\ \leftarrow \text{token 1: item } C \end{matrix}$$

Target:

$$y = 1 \quad (\textit{token 0} < \textit{token 1})$$

Ordinal relations: non-adjacent pair

Inputs (one-hot encoded items, latent rank $A < B < C < D < E$):

$$X = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \leftarrow \text{token 0: item } D \\ \leftarrow \text{token 1: item } B \end{matrix}$$

Target:

$$y = 0 \quad (\textit{token 0} > \textit{token 1})$$